MODULE 01

Introduction to Data Science

DATA SCIENCE · LEARNING NOTES

Mohan Babu University · Course-aligned study guide

Concepts • examples • practice • revision

01. Skills for data science

Learning focus. This section introduces skills for data science as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to skills for data science. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

02. Tools for data science

Learning focus. This section introduces tools for data science as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to tools for data science. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

03. Data types and collections

Learning focus. This section introduces data types and collections as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to data types and collections. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

04. Data preprocessing

Learning focus. This section introduces data preprocessing as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to data preprocessing. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

05. Data analysis and analytics

Learning focus. This section introduces data analysis and analytics as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to data analysis and analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

06. Descriptive analytics

Learning focus. This section introduces descriptive analytics as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to descriptive analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

07. Diagnostic analytics

Learning focus. This section introduces diagnostic analytics as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to diagnostic analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

08. Predictive analytics

Learning focus. This section introduces predictive analytics as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to predictive analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

09. Prescriptive analytics

Learning focus. This section introduces prescriptive analytics as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to prescriptive analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

10. Exploratory analysis

Learning focus. This section introduces exploratory analysis as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to exploratory analysis. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

11. Mechanistic analysis

Learning focus. This section introduces mechanistic analysis as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to mechanistic analysis. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

12. Ethics and privacy

Learning focus. This section introduces ethics and privacy as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to ethics and privacy. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

13. A practical workflow

Learning focus. This section introduces a practical workflow as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to a practical workflow. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

14. Module review

Learning focus. This section introduces module review as part of introduction to data science. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to module review. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?